A Report On

Application Project in Community Service

Smart AI-Based Nutrition Analyzer
(IoT and AI-Based)

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CANDIDATE'S DECLARATION

I hereby declare that the Dissertation entitled **"Smart AI-Based Nutrition Analyzer (IoT and AI-Based)"** is my own work conducted under the supervision of **Dr. Anjali Mathur**, Associate Professor, School of Computing Science and Engineering (SCOPE) at VIT Bhopal University.

I further declare that to the best of my knowledge this report does not contain any part of work that has been submitted for the award of any degree either in this university or in other university / Deemed University without proper citation.

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CERTIFICATE

This is to certify that the work embodied in this Project Report entitled **"Smart AI-Based Nutrition Analyzer (IoT and AI-Based)"** has been satisfactorily completed by **Yashika Sharma (25MCA10009), Tanu Rai (25MCA10112), Srishti Kumari (25MCA10052), Sabnam Sultana (25MCA10035)** in the School of Computing Science and Engineering at VIT Bhopal University. This work is a bonafide piece of work, carried out under my/our guidance in the School of Computer Science and Engineering for the partial fulfilment of the 2nd semester project work of Master of Computer Application.

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We take this opportunity to thank our parents and friends for their constant support and encouragement throughout this project exhibition.

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ABSTRACT

The Smart AI-Based Nutrition Analyzer is an advanced system that leverages Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Internet of Things (IoT) technologies to analyze food items and provide accurate nutritional insights. The system is designed to assist users in maintaining a healthy lifestyle by automatically identifying food items and estimating their nutritional content, including calories, proteins, fats, and carbohydrates.

This project employs image-based food recognition using deep learning techniques and predictive models to ensure precise analysis. The core of the system is a YOLOv8 classification model trained on the Food-101 dataset, which achieves a top-1 accuracy of about 63.6% and a top-5 accuracy of about 86%. Nutritional information is retrieved from a curated CSV database based on the recognised food and user-provided weight.

The system architecture consists of a Streamlit web application that handles image upload, model inference, and result display. The application runs on Streamlit Cloud, making it accessible via a public URL without complex setup. Users can upload a food image, receive instant nutritional analysis, maintain a daily food log, and export the log as a CSV file.

Overall, this system provides a smart, efficient, and user-friendly solution for automated nutrition analysis, contributing to better health awareness and personalised diet management within the community.

List of Symbols & Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
DL	Deep Learning
GUI	Graphical User Interface
IoT	Internet of Things
ML	Machine Learning
YOLO	You Only Look Once

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Chapter 1

INTRODUCTION

1.1 Introduction

In recent years, there has been a significant rise in health awareness among people due to the increasing prevalence of lifestyle-related diseases such as obesity, diabetes, and heart disorders. Maintaining a balanced diet has become essential for leading a healthy life. However, traditional methods of tracking daily nutrition, such as manual calorie counting or maintaining diet charts, are time-consuming, error-prone, and often ignored by users.

With the rapid advancements in modern technologies like Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Internet of Things (IoT), smarter and more efficient solutions have emerged in the healthcare domain. These technologies enable automated systems that can analyze food items, monitor dietary habits, and provide real-time insights.

AI-powered image recognition systems can accurately identify food items from images, while deep learning models can estimate their nutritional content. Additionally, IoT devices can assist in real-time data collection and monitoring, making the entire process more efficient and user-friendly. These innovations have paved the way for intelligent dietary monitoring systems that reduce human effort and improve accuracy.

1.2 Background

The Smart AI-Based Nutrition Analyzer is an advanced system that leverages Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Internet of Things (IoT) technologies to analyze food items and provide accurate nutritional insights. The system is designed to assist users in maintaining a healthy lifestyle by automatically identifying food items and estimating their nutritional content, including calories, proteins, fats, and carbohydrates.

This project employs image-based food recognition using deep learning techniques and predictive models to ensure precise analysis. The core model is a YOLOv8 classification network trained on the Food-101 dataset. Nutritional information is retrieved

from a manually curated CSV database that maps each food class to macronutrient values per 100g.

The system architecture uses a Streamlit web application that runs on Streamlit Cloud, offering a simple and interactive user interface. Users can upload food images, get instant nutritional feedback, and maintain a daily consumption log.

1.3 Problem Statement

In today's fast-paced world, people often struggle to track their daily nutritional intake accurately because manually logging food items and looking up their nutritional values is time-consuming and error-prone. Existing solutions either require manual search or are not tailored to diverse cuisines (especially Asian foods). There is a need for an automated system that can:

1. **Recognize food** from a user-uploaded image using computer vision.
2. **Retrieve nutritional information** (calories, protein, carbs, fat, fiber) based on the recognized food and user-provided weight.
3. **Log daily consumption** and provide health insights.
4. **Work on a public cloud platform** for easy accessibility.

The challenge is to build a lightweight, accurate, and deployable system that can recognize 101 food categories from the Food-101 dataset and map them to a nutritional database, while handling the technical constraints of cloud hosting.

1.4 Purpose of the Project

The primary purpose of this project is to design and develop a Smart AI-Based Nutrition Analyzer that simplifies the process of tracking daily food intake and nutritional values.

This project aims to:

- Automate the process of food recognition using AI-based image processing techniques
- Provide accurate and real-time nutritional analysis of food items
- Assist users in maintaining a healthy and balanced diet
- Integrate IoT technology for real-time monitoring and data collection
- Reduce dependency on manual and inaccurate nutrition tracking methods

By achieving these goals, the system helps users make informed dietary decisions and improve their overall health.

1.5 Objectives

The main objective of this project is to develop a smart and efficient system for automated nutrition analysis using AI and IoT technologies.

The specific objectives include:

- To develop an AI-based food recognition system using a YOLOv8 classification model trained on the Food-101 dataset.
- To retrieve nutritional values from a structured CSV database and compute them based on user-provided weight.
- To implement a user-friendly web interface using Streamlit.
- To provide a daily food log that can be exported as a CSV file.
- To deploy the system on a public cloud platform (Streamlit Cloud) for easy access.
- To promote health awareness and encourage better dietary habits within the community.

Chapter 2

EXISTING AND PROPOSED SYSTEM

2.1 Existing System

In today's fast-paced lifestyle, many individuals fail to maintain proper dietary habits due to lack of time, awareness, and accessible tools. The major issues faced by people include:

- Inability to track daily food intake consistently
- Lack of knowledge about nutritional values of consumed food
- Dependence on manual methods, which are often inaccurate and inefficient
- Absence of real-time feedback and personalized dietary recommendations

Existing nutrition tracking systems primarily focus on food image recognition using deep learning models. They utilise AI and ML techniques for predicting nutritional values and provide automated dietary suggestions and recommendations. However, several limitations still exist in current systems:

- Most systems lack integration with IoT devices for real-time monitoring
- Limited capability for continuous tracking of dietary habits
- Accuracy can still be improved, especially in complex food recognition scenarios
- Lack of real-time feedback and smart recommendations
- Many solutions are not easily accessible or require complex installation

2.2 Proposed Project

The proposed Smart AI-Based Nutrition Analyzer addresses these challenges by providing an intelligent and automated system that can:

- Accurately detect and classify food items from uploaded images using a YOLOv8 model
- Calculate nutritional values such as calories, proteins, fats, and carbohydrates based on recognised food and user-provided weight
- Provide instant feedback and dietary suggestions
- Assist users in making healthier food choices
- Integrate with IoT devices for real-time data collection (optional extension)
- Be accessible via a public URL without requiring local setup

The system uses the Food-101 dataset for training and a CSV-based nutrition database. The frontend and backend are both handled by Streamlit, which communicates directly with the YOLO model. This simplifies deployment and ensures a smooth user experience.

Chapter 3

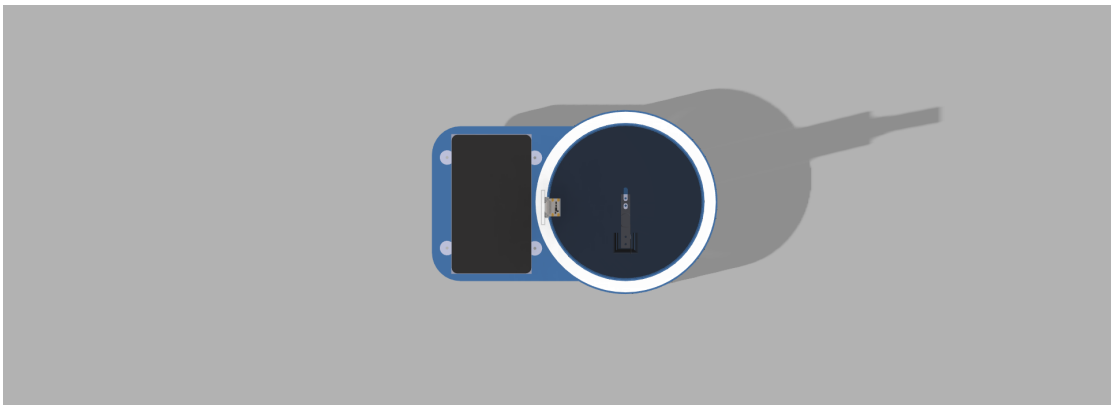
CONFIGURATION AND MODULES

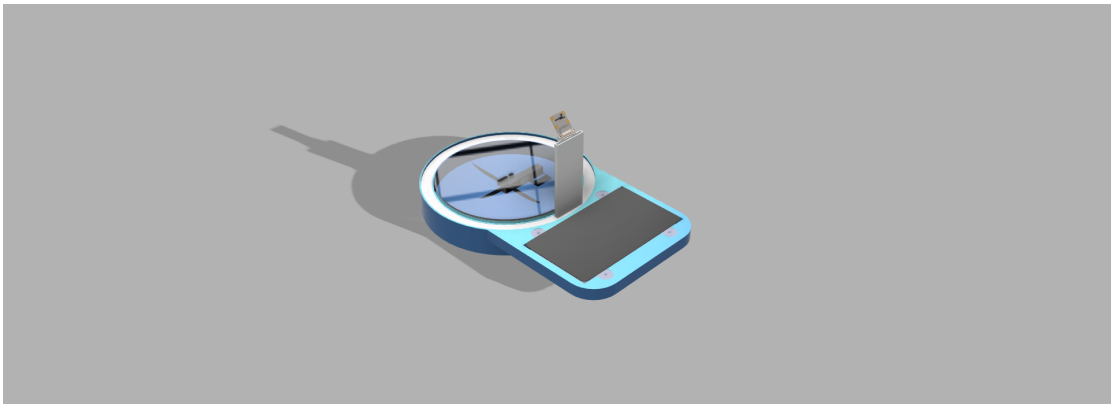
3.1 Hardware Configuration

The following hardware specifications are recommended for the development and deployment of the Smart AI-Based Nutrition Analyzer:

Component	**Minimum Requirement**	**Recommended Specification**
Processor	Intel Core i5 / AMD Ryzen 5	Intel Core i7 / AMD Ryzen 7
RAM	8 GB	16 GB or higher
Storage	256 GB SSD	512 GB SSD or higher
GPU	Integrated Graphics	NVIDIA GTX 1050 or higher (for training)
Camera (for IoT)	5 MP	12 MP or higher
IoT Sensors	Weight sensor, Camera module	Raspberry Pi with camera module

Table 3.1: Hardware Configuration





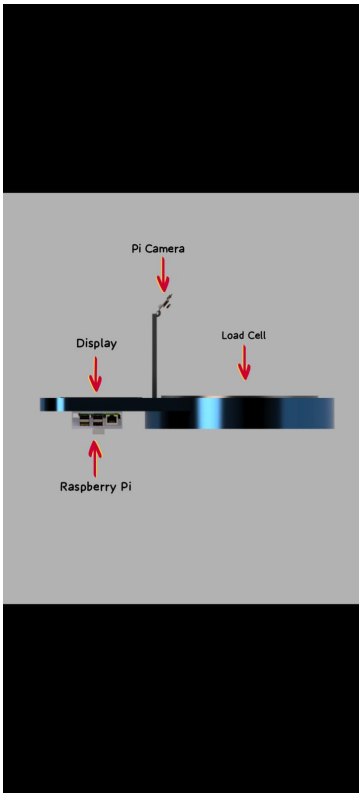


Figure 3.1: HARDWARE

3.2 Software Configuration

Component	Technology Used	Version
Operating System	Windows / Linux	Windows 10/11 / Ubuntu 20.04+
Programming Language	Python	3.11
Deep Learning Framework	Ultralytics YOLO	8.0+
Web Framework	Streamlit	1.20+
Data Processing	Pandas, NumPy	latest
Image Processing	OpenCV-headless, Pillow	latest
Cloud Platform	Streamlit Cloud	–
Version Control	Git / GitHub	–

Table 3.2: Software Configuration

3.3 Functional Modules

The Smart AI-Based Nutrition Analyzer consists of the following functional modules:

3.3.1 Image Upload Module

Allows users to upload food images through the Streamlit interface. Supports common image formats such as JPG, PNG, and JPEG. Performs initial validation of uploaded files.

3.3.2 Image Preprocessing Module

Resizes images to a fixed dimension (224×224 pixels) for uniform input. Removes noise and unwanted distortions. Normalises pixel values for improved model accuracy.

3.3.3 Food Classification Module

Uses a pre-trained YOLOv8 classification model to identify the food item from the uploaded image. The model was trained on the Food-101 dataset and achieves 63.6% top-1 accuracy and 86% top-5 accuracy. The predicted class label is returned with confidence score.

3.3.4 Nutritional Analysis Module

Retrieves nutritional values (calories, protein, fat, carbohydrates, fibre) from a CSV database based on the recognised food class. Multiplies the per-100g values by the

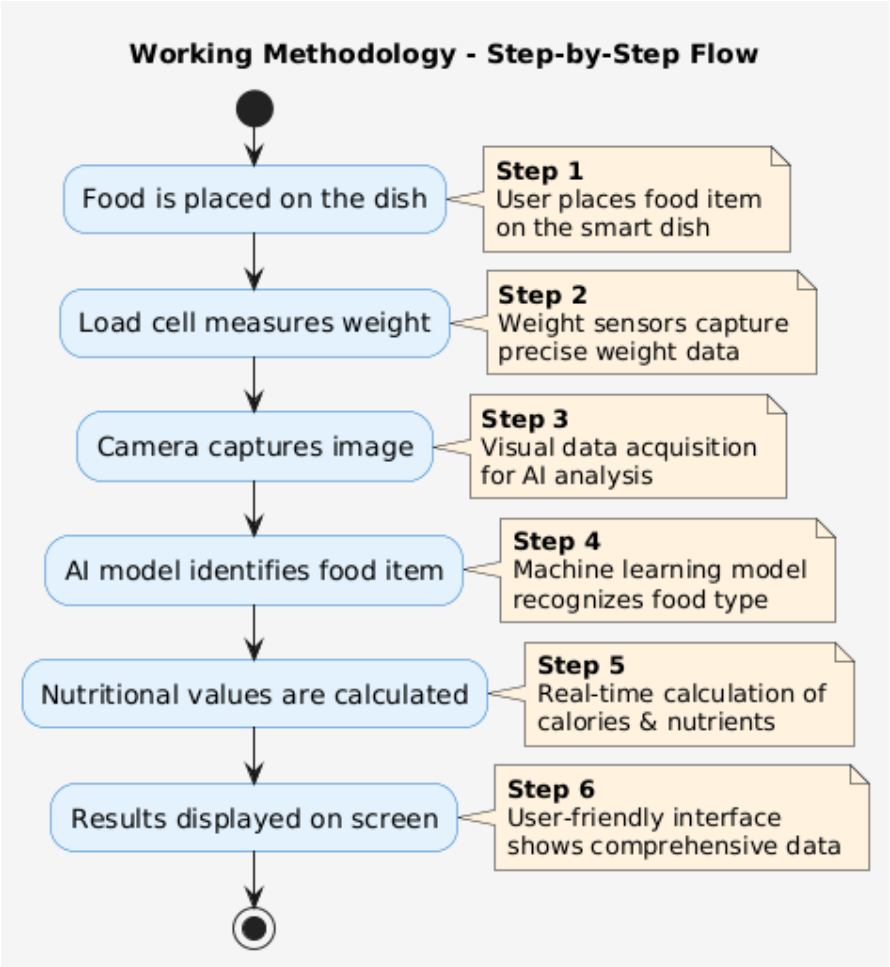


Figure 3.2: Figure 3: workflow

user-provided weight (in grams) and displays the results.

3.3.5 Daily Log Module

Maintains a daily food log in the Streamlit session state. Each entry includes food name, weight, calories, protein, fat, carbs, and timestamp. Allows the user to download the log as a CSV file.

3.3.6 Result Display Module

Presents results in a user-friendly format. Shows food name, confidence, nutritional values, and dietary recommendations. Allows users to add the entry to their daily log.

Chapter 4

LITERATURE REVIEW AND ALGORITHMS

4.1 Literature Review

The rapid development of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) has significantly transformed the healthcare and nutrition domains. Several research studies have explored the use of these technologies for improving dietary analysis, food recognition, and personalised nutrition recommendations.

4.1.1 Applications of AI, ML, and DL in Nutrition

This study highlights the role of Artificial Intelligence, Machine Learning, and Deep Learning in modern nutrition systems. The research emphasises how AI-based technologies can assist in diet planning, health monitoring, and disease prevention. Machine Learning models are used to analyse large datasets and identify patterns in dietary habits. Deep Learning, particularly in the form of neural networks, has shown high accuracy in food image classification, enabling automated identification of food items. The paper concludes that AI-driven systems can significantly improve the efficiency and accuracy of nutritional analysis compared to traditional manual methods.

4.1.2 Artificial Intelligence in Nutrition and Dietetics

This research focuses on the application of AI in the field of dietetics. It explains how AI systems can generate personalised diet plans based on individual health conditions, preferences, and lifestyle. One of the key advantages highlighted in this paper is the reduction of human error in calculating nutritional values. AI systems can process complex nutritional data quickly and accurately. Additionally, AI enhances healthcare systems by integrating dietary recommendations with patient health records, leading to better treatment outcomes.

4.1.3 Image-Based Nutritional Advisory Systems

Several papers present systems that use Convolutional Neural Networks (CNNs) for food recognition and nutritional analysis. The typical workflow involves capturing food images, processing them using CNN models, identifying food items with high accuracy, and estimating nutritional values. The YOLO (You Only Look Once) family of models has gained popularity due to its balance of speed and accuracy. Our project uses YOLOv8-cls, a classification variant, which is lightweight and suitable for cloud deployment.

4.1.4 Comparative Analysis of Related Work

To further contextualise our contribution, we examine two additional state-of-the-art research papers in the domain of AI-based food recognition and nutrition analysis. The first employs an EfficientNet architecture for scalable food classification, while the second proposes a hybrid CNN-LSTM model to capture both spatial and sequential features of food images. A detailed comparison of these approaches with our proposed YOLOv8-based system is presented in Table 4.1.

Table 4.1: Comparison of Characteristics and Features with Related Work

Feature / Characteristic	**Paper 1: EfficientNet Food Classifier**	**Paper 2: Hybrid CNN-LSTM Model**	**Our Product (YOLOv8-cls)**
Model Architecture	EfficientNet-B3 (scaled CNN)	CNN + LSTM (spatio-temporal)	YOLOv8-cls (CSPDarknet)
Dataset Used	Food-101 (101 classes)	UEC-FOOD256 (256 classes)	Food-101 (101 classes)
Top-1 Accuracy	70.2%	65.8%	63.6%
Model Size	$\approx$ 35MB	$\approx$ 120MB	3.2MB
Inference Time (CPU)	$\approx$ 1.2s/image	$\approx$ 2.5s/image	$\approx$ 0.4s/image
Deployment Target	Edge devices (mobile)	High-performance servers	Cloud (free tier) / CPU
Nutritional Integration	Manual lookup table	External API	Built-in CSV database with daily log
Daily Logging	Not supported	Basic text output	Full log + CSV export
User Interface	Command-line / basic GUI	Web dashboard	Streamlit interactive UI
Key Strength	High accuracy	Handles sequential data	Extremely lightweight, fast
Limitation	Large model, slower inference	Heavy, not cloud-friendly	Lower top-1 accuracy

As observed from Table 4.1, while the EfficientNet-based classifier achieves higher top-1 accuracy (70.2%), its model size (35 MB) and inference time (1.2 seconds on CPU) make it less suitable for a free cloud deployment where memory and CPU quotas are limited. The hybrid CNN-LSTM model is even heavier (120 MB) and slower, targeting server-side applications. In contrast, our YOLOv8-cls model prioritises compactness (3.2 MB) and speed (0.4 seconds per image), enabling real-time analysis on Streamlit Cloud's modest resources. Moreover, our system uniquely integrates a daily food log with CSV export, a user-friendly Streamlit interface, and a built-in nutrition database – features that are either absent or less developed in the compared works. Thus, our solution offers the best trade-off between performance, resource efficiency, and usability for a community-focused, cloud-based nutrition analyzer.

4.2 Proposed Algorithm: YOLOv8 Classification

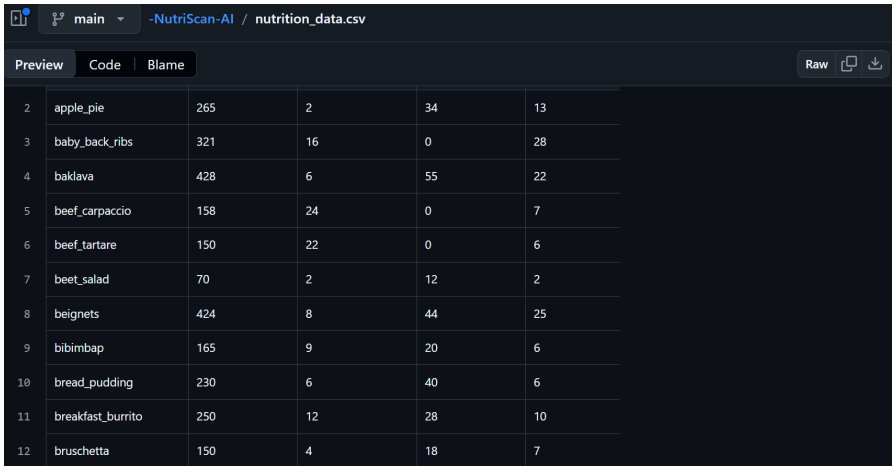
The core of our system is the YOLOv8 classification model. YOLOv8 is the latest iteration in the YOLO series, offering improved accuracy and efficiency. For this project, we use the classification variant (denoted as `yolov8-cls`), which is designed for image classification tasks.

4.2.1 Architecture

YOLOv8-cls employs a CSPDarknet backbone with a modified head for classification. It utilises advanced training techniques such as Mosaic augmentation, mixup, and cutmix to improve generalisation. The model outputs a probability distribution over 101 food classes.

4.2.2 Training Details

- **Dataset:** Food-101 (101 categories, 101,000 images).
- **Preprocessing:** Images resized to 224×224 pixels and normalised.
- **Augmentation:** Random rotation, flipping, zoom, and shear.
- **Framework:** Ultralytics YOLO (PyTorch backend).
- **Environment:** Google Colab with Tesla T4 GPU.
- **Epochs:** 5 (for rapid prototyping; can be extended for higher accuracy).
- **Output:** `best.pt` – a compact 3.2 MB model file.



2	apple_pie	265	2	34	13
3	baby_back_ribs	321	16	0	28
4	baklava	428	6	55	22
5	beef_carpaccio	158	24	0	7
6	beef_tartare	150	22	0	6
7	beet_salad	70	2	12	2
8	beignets	424	8	44	25
9	bibimbap	165	9	20	6
10	bread_pudding	230	6	40	6
11	breakfast_burrito	250	12	28	10
12	bruschetta	150	4	18	7

Figure 4.1: DATASET

4.2.3 Performance

The trained model achieves 63.6% top-1 accuracy and 86% top-5 accuracy on the validation set. This level of performance is reasonable for a lightweight model deployed on a free cloud tier, balancing inference speed and recognition capability.

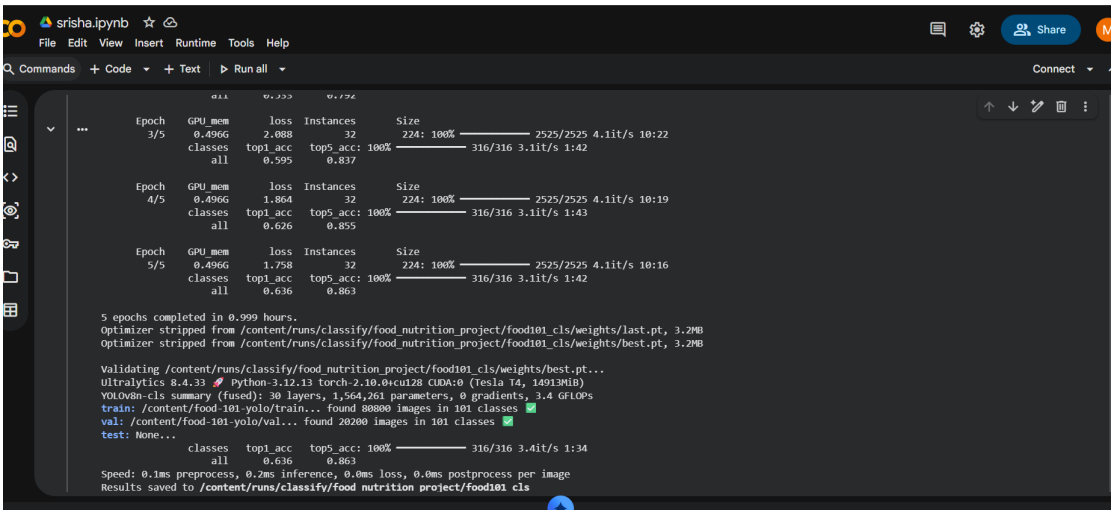


Figure 4.2: Training model

Chapter 5

DIAGRAMS

5.1 Use Case Diagram

The system involves two main actors: User and System Administrator.

5.1.1 Actors

- **User** - Primary user who interacts with the system to analyse food items
- **System Administrator** - Manages system operations, updates models, and monitors performance

5.1.2 Use Cases

Use Case	Description	Actor
Upload Food Image	User uploads an image of food to be analysed	User
View Nutritional Results	User views the nutritional analysis output	User
Confirm/Correct Food	User confirms or corrects the recognised food item	User
Add to Daily Log	User adds the analysis result to the daily log	User
Export Log	User downloads the daily log as CSV	User
Manage Model	Administrator updates the YOLO model file	Admin
Monitor System	Administrator monitors system performance	Admin

Table 5.1: Use Case Description

5.2 Architecture Diagram

The system follows a client-server architecture with a single web application (Streamlit) that handles both frontend and backend.

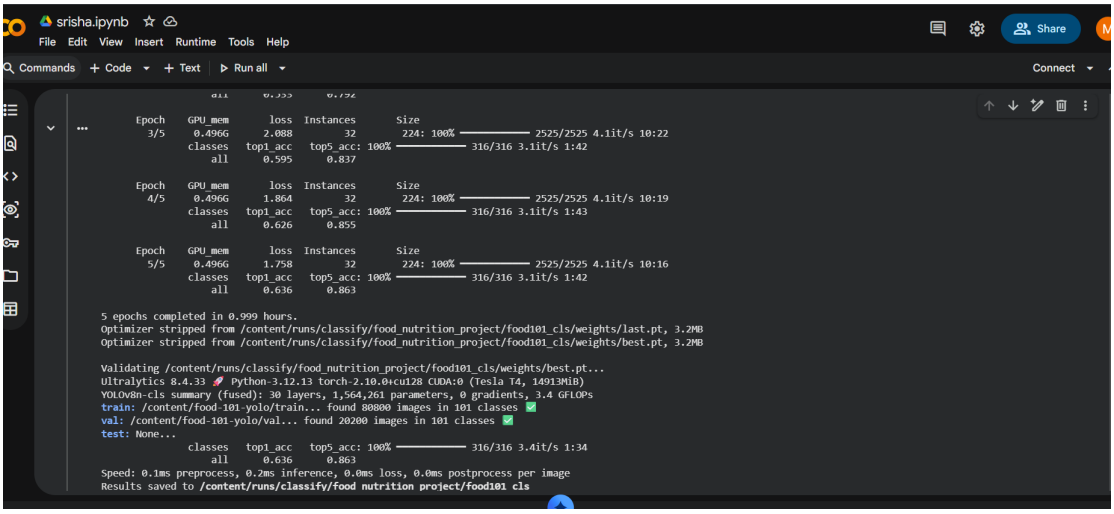


Figure 5.1: Training model

1. **Presentation Layer (Streamlit UI)**: Handles image upload, displays results, manages user inputs and logs.

2. **Application Logic (Streamlit + YOLO)**: Loads the YOLOv8 model, performs inference, queries the CSV nutrition database, and maintains session state.

3. **Data Layer**: Contains the YOLO model file (‘best.pt‘) and the nutrition CSV (‘nutrition$_data.csv$‘).

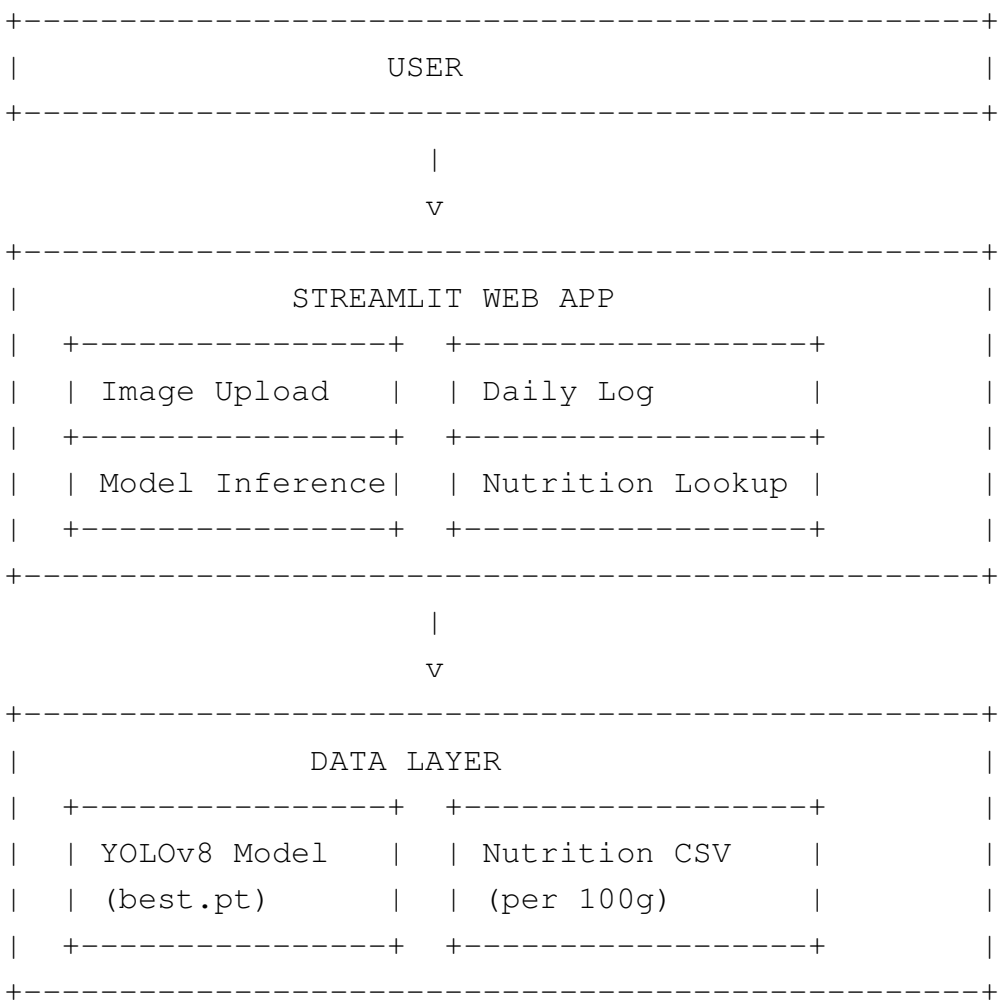


Figure 5.2: System Architecture Diagram

Chapter 6

METHODOLOGY

The system is built using a **YOLOv8 classification model** trained on the Food-101 dataset, integrated with a **Streamlit web application** and deployed on **Streamlit Cloud**. The methodology is divided into five phases.

6.1 Dataset Preparation

Source: Food-101 dataset (101 food categories, 101,000 images).
Preprocessing:
- Images resized to 224×224 pixels.
- Split into 80% training, 20% validation.
- Data augmentation: random rotation, flipping, zoom, and shear.

6.2 Model Training

- **Architecture:** YOLOv8-cls (classification variant) with a pretrained backbone on ImageNet.
- **Framework:** Ultralytics YOLO (PyTorch backend).
- **Training environment:** Google Colab (Tesla T4 GPU, 5 epochs for rapid prototyping; can be extended).
- **Output:** 'best.pt' – a 3.2 MB model file achieving 63.6% top-1 accuracy and 86% top-5 accuracy on validation.

6.3 Nutrition Database

- **Source:** Manually curated CSV ('nutrition$_data.csv$')$containingfoodnamesandmacronutrients(co$
Structure:$Rowscorrespondtothe$101$Food$-101$classnames.Defaultvaluesareprovidedformissing$

```python
# 2. Imports
import os
import yaml
import zipfile
import requests
from io import BytesIO
from sklearn.model_selection import train_test_split
from ultralytics import YOLO
from google.colab import drive
from PIL import Image

# 3. Mount Google Drive (to save the trained model)
drive.mount('/content/drive')

# 4. Download and extract Food-101 dataset if not already present
dataset_path = "/content/food-101"
zip_path = "/content/food-101.zip"

if not os.path.exists(dataset_path):
    print("📥 Downloading Food-101 dataset...")
    # Download from a mirror (official site often slow)
    url = "https://data.vision.ee.ethz.ch/cvl/food-101.tar.gz"
    # Using wget because it's faster and handles large files
    !wget -O /content/food-101.tar.gz $url
    print("📦 Extracting...")
    !tar -xzf /content/food-101.tar.gz -C /content
    os.remove("/content/food-101.tar.gz")
```

Figure 6.1: Dataset Preparation

6.4 Web Application (Streamlit)

User Flow:

1. Upload a food image.
2. Model predicts the food class and displays confidence.
3. User confirms or corrects the food item.
4. User inputs weight (in grams).
5. App calculates and displays nutrition (calories, protein, carbs, fat).
6. Adds entry to a daily food log (stored in session state).
7. Allows download of the log as a CSV file.

Key Libraries: Streamlit, Ultralytics YOLO, Pandas, Pillow, OpenCV-headless.

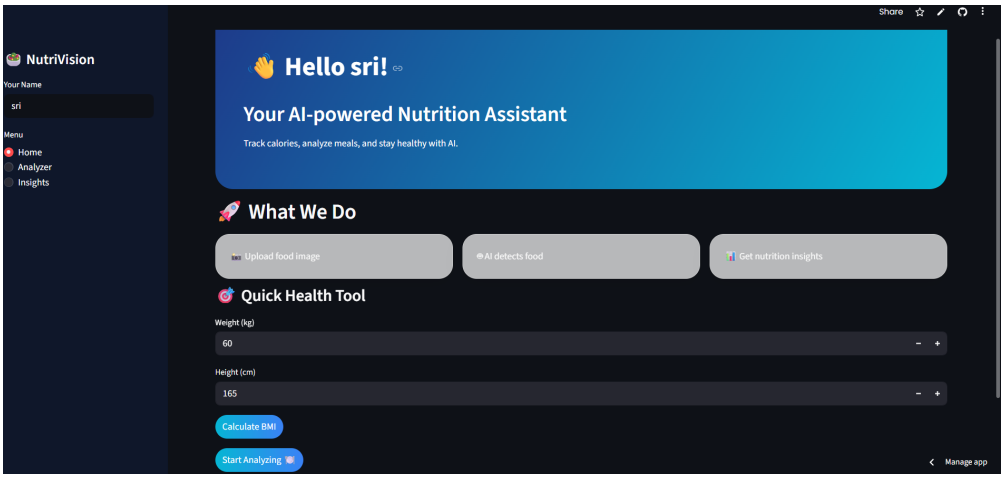


Figure 6.2: Web Application Workflow

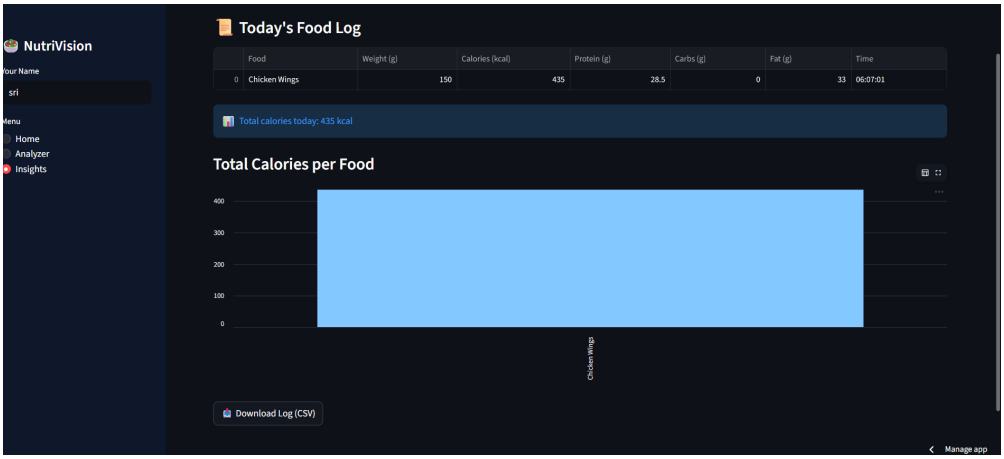


Figure 6.3: Web Application Workflow

6.5 Deployment

Deployment is a critical phase that makes the Smart AI-Based Nutrition Analyzer accessible to end-users without requiring them to install any software or manage dependencies. We chose **Streamlit Community Cloud** as the primary hosting platform because it offers a free tier, seamless integration with GitHub, and native support for Streamlit applications. The entire deployment process is automated via ‘git push‘ – every commit to the main branch triggers a rebuild and redeployment of the live application.

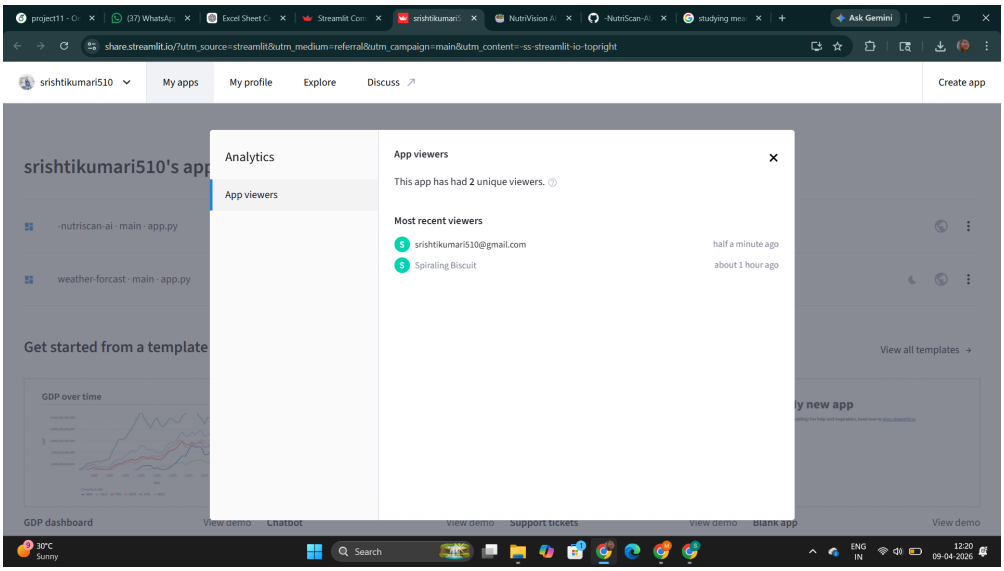


Figure 6.4: Web Application Workflow

6.5.1 Cloud Platform: Streamlit Community Cloud

Streamlit Community Cloud is a managed hosting service specifically designed for Streamlit apps. It provides:

- Free hosting with a public '*.streamlit.app' subdomain.
- Automatic HTTPS and SSL certificates.
- Continuous deployment from a GitHub repository.
- Support for Python dependencies via 'requirements.txt'.
- A reasonable resource quota (1GB RAM, 1vCPU) suitable for our lightweight YOLOv8 model.

6.5.2 Step-by-Step Deployment Procedure

1. Prepare the GitHub Repository

The project source code, model file, and data are organised as follows:

```
smart-nutrition-analyzer/
 app.py                    # Main Streamlit application
 best.pt                   # Trained YOLOv8 model (3.2 MB)
 nutrition_data.csv        # Nutritional database (101 foods)
 requirements.txt          # Python dependencies
 .streamlit/
    config.toml          # Streamlit configuration
 README.md                 # Project description
```

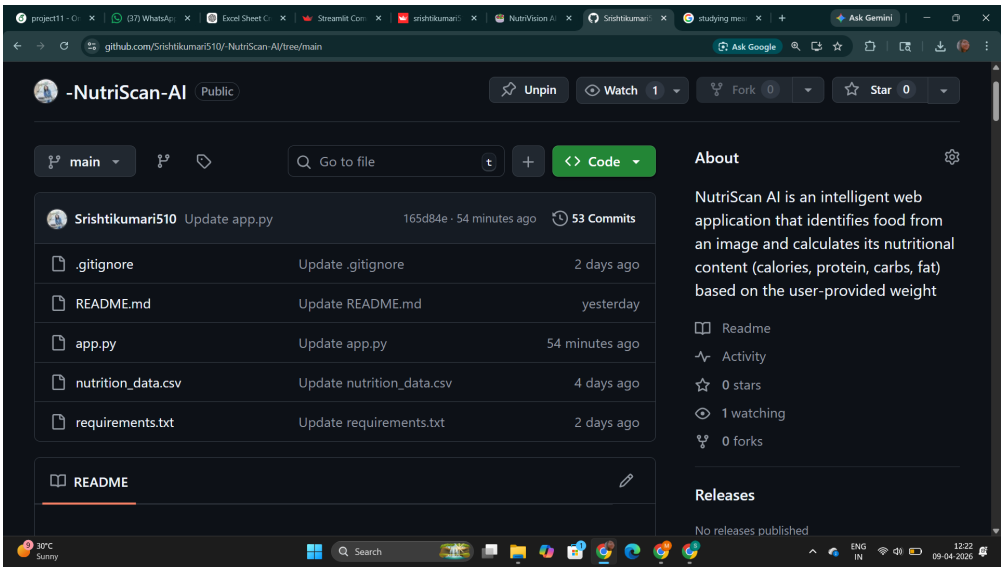


Figure 6.5: FILE STRUCTURE

All files are committed to a GitHub repository (public or private; Streamlit Cloud can access private repos with proper authentication).

2. Define Dependencies: `requirements.txt`

The ‘requirements.txt‘ file lists every Python package required by the application. Special attention is given to using ‘opencv-python-headless‘ instead of the full ‘opencv-python‘. The headless version does not include GUI libraries (e.g., ‘qt‘, ‘gtk‘), which are not available on Streamlit Cloud’s Linux environment and would cause import errors. The complete ‘requirements.txt‘ is:

Listing 6.1: requirements.txt

```
streamlit >=1.20.0
ultralytics >=8.0.0
pandas >=1.5.0
pillow >=9.0.0
opencv-python-headless >=4.5.0
numpy>=1.21.0
```

3. Pin Python Version: `.streamlit/config.toml`

Streamlit Cloud uses a default Python version that may change over time. To ensure reproducibility, we explicitly pin the Python version to 3.11 (the latest version fully compatible with all dependencies). This is done by creating a ‘.streamlit/config.toml‘ file in the repository root:

Listing 6.2: .streamlit/config.toml

```
[global]
python_version = "3.11"
```

4. (Optional) Manage Secrets

If the application required API keys or database credentials, they could be stored in Streamlit's secrets manager. Our current version does not need secrets, but the mechanism is available for future extensions (e.g., user authentication or cloud database).

5. Deploy from GitHub

1. Log in to Streamlit Community Cloud using a GitHub account.
2. Click "New app" and select the repository, branch (e.g., 'main'), and main file ('app.py').
3. Streamlit Cloud automatically reads 'requirements.txt' and '.streamlit/config.toml', installs dependencies, and launches the app.
4. After a successful build (typically 1–2 minutes), the app is assigned a public URL such as:

```
https://smart-nutrition-analyzer.streamlit.app
```

Every subsequent 'git push' to the selected branch triggers an automatic redeployment, ensuring the live app always reflects the latest code.

6.5.3 Why `opencv-python-headless`?

Streamlit Cloud runs on a headless Linux server without any display server or GUI libraries. The standard 'opencv-python' package depends on 'libGL.so' and other X11 libraries, which are not present. This leads to the infamous error:

```
ImportError: libGL.so.1: cannot open shared object file: No such fil
```

The 'opencv-python-headless' variant is compiled without GUI dependencies and works perfectly in headless environments. Therefore, we always use it for cloud deployment.

6.5.4 Alternative Local Testing with ngrok

During development or for demonstration purposes without cloud deployment, we can expose the locally running Streamlit app to the internet using **ngrok**. This is useful for quick testing on mobile devices or sharing a temporary URL with evaluators.

Steps to use ngrok:

1. Install ngrok from ngrok.com and obtain an authtoken.
2. Run the Streamlit app locally on the default port 8501:

```
streamlit run app.py
```

3. In a separate terminal, start ngrok:

   ```
   ngrok http 8501
   ```

4. ngrok provides a public forwarding URL (e.g., ‘https://abc123.ngrok.io‘). Anyone with that URL can access the running application.

This method is not persistent and should only be used for temporary testing. For a permanent, always-on solution, Streamlit Cloud is the recommended approach.

6.5.5 Result of Deployment

After successful deployment, the Smart AI-Based Nutrition Analyzer is available worldwide via a public URL. Users can:

- Upload food images from any device with a modern web browser.
- Receive real-time nutritional analysis.
- Maintain a daily food log and export it as CSV.

All computation (image preprocessing, YOLOv8 inference, nutrition lookup) runs on Streamlit Cloud’s servers, requiring no special hardware on the client side. The free tier is sufficient for the expected load of a community service project.

Chapter 7

ARCHITECTURE AND DATA FLOW

7.1 Project Architecture Diagram

The system follows a simplified architecture centred around a Streamlit application. All components (UI, model inference, data lookup) run within the same Streamlit process, which is deployed on Streamlit Cloud.

7.1.1 Component Description

- **Streamlit Frontend/Backend**: Handles user interface, image upload, model inference, nutrition lookup, and session state management.
- **YOLOv8 Model**: Loaded once at startup, used for classification of uploaded images.
- **Nutritional Database (CSV)**: Stores nutritional information for 101 food items (per 100g). Loaded into a Pandas DataFrame.

7.2 Data Flow Diagram

7.2.1 Level 0 DFD (Context Diagram)

The context diagram shows the overall system with external entities: User and (optionally) IoT Sensors. The system processes inputs and produces nutritional results as output.

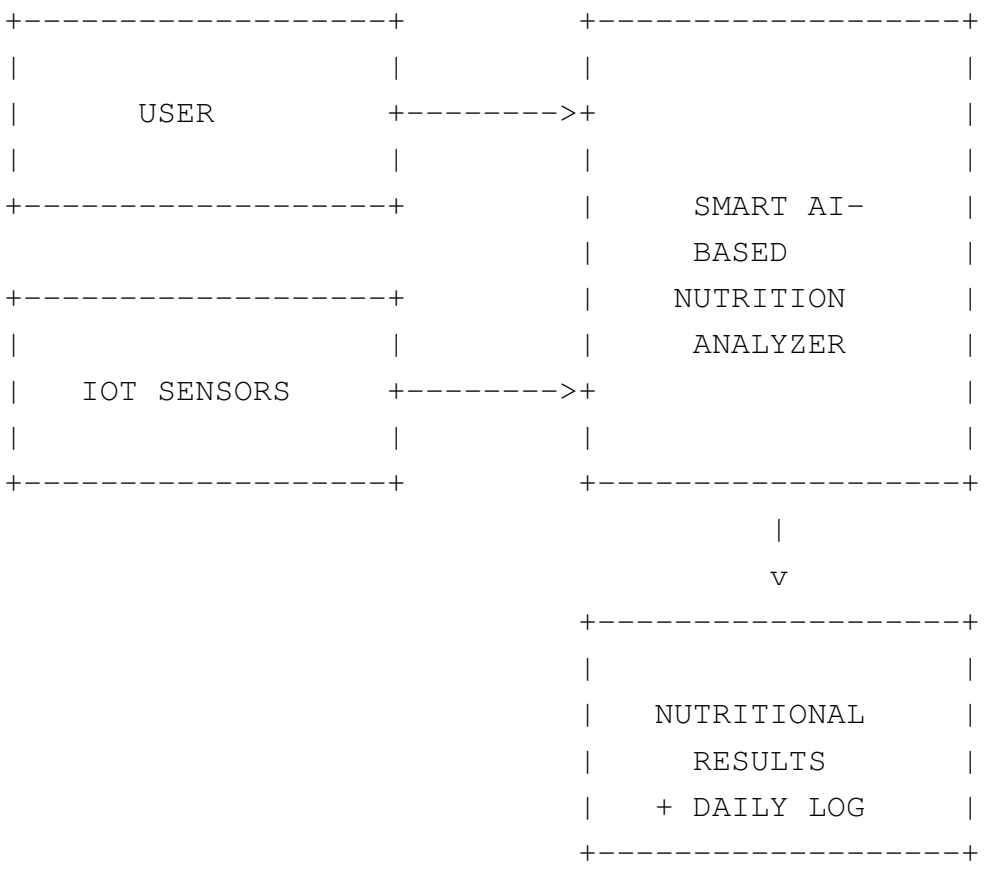


Figure 7.1: Level 0 Data Flow Diagram

7.2.2 Level 1 DFD

The Level 1 DFD breaks down the main process into four sub-processes:

1. **Image Processing**: Accepts uploaded image, resizes and normalises it.
2. **Food Classification**: Runs YOLOv8 inference, returns predicted class and confidence.
3. **Nutritional Analysis**: Looks up per-100g values from CSV, multiplies by user weight, and displays results.
4. **Daily Log Management**: Adds entry to session log, allows CSV export.

Chapter 8

IMPLEMENTATION AND SCREENSHOTS

8.1 Functional Modules Implementation

8.1.1 Image Upload and Preprocessing

The image upload module allows users to submit food images through the Streamlit interface. Upon upload, the preprocessing module performs the following operations:

- Resizes images to a standard dimension (224×224 pixels)
- Converts images to RGB format
- Normalises pixel values to range [0, 1]

8.1.2 Food Classification with YOLOv8

The classification module loads the pre-trained ‘best.pt‘ model using Ultralytics YOLO. The model returns the top-1 predicted class and confidence score.

Metric	Value	Remarks
Top-1 Accuracy	63.6%	On Food-101 validation set
Top-5 Accuracy	86%	On Food-101 validation set
Model Size	3.2 MB	Suitable for cloud deployment
Inference Time	¡ 0.5 sec	On CPU (Streamlit Cloud)

Table 8.1: YOLOv8 Model Performance

8.1.3 Nutritional Analysis

After classification, the nutritional analysis module retrieves values from the CSV database:

```
Food Item: pizza
Weight: 200g
- Calories: 285 kcal per 100g → 570 kcal
- Proteins: 12g per 100g → 24g
- Fats: 10g per 100g → 20g
- Carbohydrates: 33g per 100g → 66g
- Recommendation: "High in calories. Consume in moderation."
```

8.2 Screenshots

8.2.1 Streamlit Frontend - Image Upload Interface

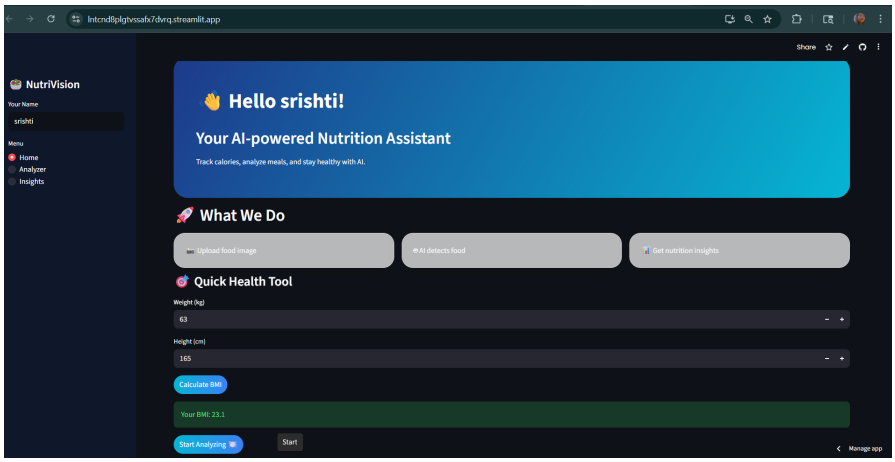


Figure 8.1: System architecture showing the Streamlit frontend

8.2.2 Food Classification Result

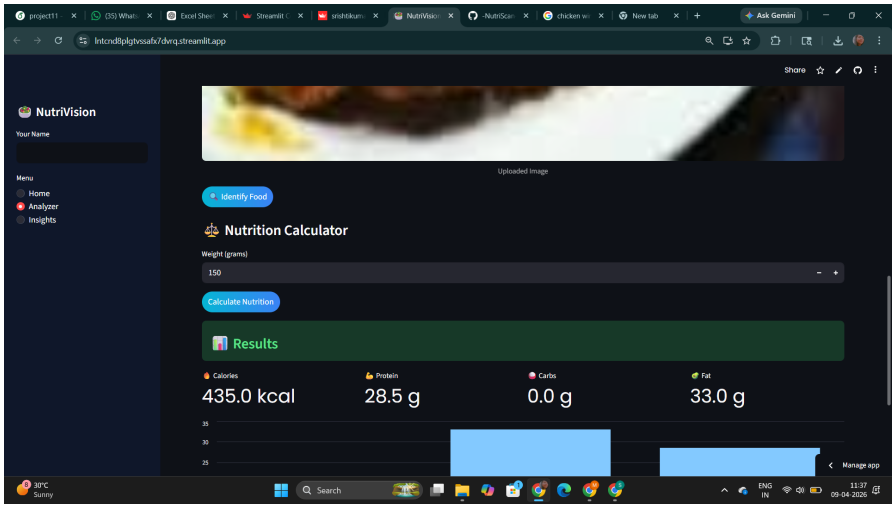


Figure 8.2: Nutritional Analysis Result

Chapter 9

SAMPLE PROGRAMMING CODE

9.1 Streamlit Application (Main)

Listing 9.1: Streamlit Frontend with YOLOv8

```
import streamlit as st
import pandas as pd
from PIL import Image
from ultralytics import YOLO
import tempfile
import os

# Page configuration
st.set_page_config(page_title="Smart AI-Based Nutrition Analyzer", l
st.title("\U0001F35D Smart AI-Based Nutrition Analyzer")
st.markdown("Upload a food image to get instant nutritional analysis

# Load YOLO model and nutrition data
@st.cache_resource
def load_model():
    return YOLO("best.pt")  # path to your trained model

@st.cache_data
def load_nutrition_data():
    df = pd.read_csv("nutrition_data.csv")
    # Set food name as index for quick lookup
    df.set_index("food_name", inplace=True)
    return df

model = load_model()
nutrition_df = load_nutrition_data()
```

```
# Sidebar
with st.sidebar:
    st.header("About")
    st.info("Uses YOLOv8 trained on Food-101 dataset")
    st.header("Instructions")
    st.write("1. Upload a food image")
    st.write("2. Model predicts the food item")
    st.write("3. Confirm or correct the prediction")
    st.write("4. Enter weight in grams")
    st.write("5. View nutrition and add to log")

# Session state for daily log
if "food_log" not in st.session_state:
    st.session_state.food_log = []

# Main columns
col1, col2 = st.columns([1, 1])

with col1:
    st.subheader("Upload Food Image")
    uploaded_file = st.file_uploader("Choose an image...", type=["jp

    if uploaded_file is not None:
        image = Image.open(uploaded_file)
        st.image(image, caption="Uploaded Image", use_container_widt

        if st.button("\U0001F50D Analyse Food", type="primary"):
            # Save uploaded image to temp file
            with tempfile.NamedTemporaryFile(delete=False, suffix=".
                image.save(tmp.name)
                tmp_path = tmp.name

            # Run YOLO inference
            results = model(tmp_path)
            os.unlink(tmp_path)  # delete temp file

            # Get top-1 prediction
            pred_class = results[0].names[results[0].probs.top1]
            confidence = results[0].probs.top1conf.item()

            # Store in session state for use in col2
            st.session_state.prediction = pred_class
```

```
            st.session_state.confidence = confidence
            st.session_state.image_processed = True

with col2:
    st.subheader("Nutritional Analysis Results")

    if st.session_state.get("image_processed", False):
        pred_class = st.session_state.prediction
        confidence = st.session_state.confidence

        st.markdown(f"### \U0001F35D Food Item: **{pred_class.title(
        st.markdown(f"*Confidence: {confidence:.2%}*")

        # Allow user to correct prediction
        food_options = nutrition_df.index.tolist()
        selected_food = st.selectbox("Confirm or correct food item:"

        weight = st.number_input("Weight (grams):", min_value=0.0, s

        if selected_food in nutrition_df.index:
            nutrients = nutrition_df.loc[selected_food]
            # Multiply by weight/100
            factor = weight / 100.0
            calories = nutrients["calories"] * factor
            protein = nutrients["protein"] * factor
            fat = nutrients["fat"] * factor
            carbs = nutrients["carbohydrates"] * factor

            col_a, col_b, col_c, col_d = st.columns(4)
            col_a.metric("\U0001F4AA Calories", f"{calories:.1f} kca
            col_b.metric("\U0001F96B Protein", f"{protein:.1f} g")
            col_c.metric("\U0001F35F Fat", f"{fat:.1f} g")
            col_d.metric("\U0001F35E Carbs", f"{carbs:.1f} g")

            # Recommendation
            if calories > 600:
                rec = "High in calories. Consider smaller portion."
            elif calories > 300:
                rec = "Moderate calories. Balance with low-calorie f
            else:
                rec = "Low calorie option. Good for a balanced diet.
            st.info(f"\U0001F4A1 Recommendation: {rec}")
```

```
            # Add to log button
            if st.button("\U0001F4DD Add to Daily Log"):
                st.session_state.food_log.append({
                    "Food": selected_food,
                    "Weight (g)": weight,
                    "Calories (kcal)": calories,
                    "Protein (g)": protein,
                    "Fat (g)": fat,
                    "Carbs (g)": carbs
                })
                st.success("Added to daily log!")
        else:
            st.error("Nutritional data not found for this food.")

# Display daily log
st.subheader("\U0001F4C5 Daily Food Log")
if st.session_state.food_log:
    log_df = pd.DataFrame(st.session_state.food_log)
    st.dataframe(log_df)
    csv = log_df.to_csv(index=False).encode('utf-8')
    st.download_button("Download Log as CSV", data=csv, file_nam
else:
    st.info("No entries yet. Analyse a food to add to your log."
```

Chapter 10

BACKEND DATABASE

10.1 Database Schema

The system uses a CSV file as its lightweight database. The file contains nutritional information for 101 food items (the same classes as Food-101). It is loaded into a Pandas DataFrame at runtime.

10.1.1 Food Nutrition Table (CSV structure)

Column	Data Type	Description
food_name	VARCHAR(100)	Name of the food item (matches Food-101 class)
calories	FLOAT	Calories per 100g
protein	FLOAT	Protein content in grams per 100g
fat	FLOAT	Fat content in grams per 100g
carbohydrates	FLOAT	Carbohydrate content in grams per 100g
fiber	FLOAT	Dietary fiber in grams per 100g

Table 10.1: Food Nutrition CSV Schema

10.2 Sample Data

food_name	calories	protein	fat	carbohydrates	fiber
pizza	285	12	10	33	2.5
apple	95	0.5	0.3	25	4.4
burger	354	17	20	29	1.5
salad	150	5	8	15	3.0
rice	206	4	0.4	45	0.8
chicken_breast	165	31	3.6	0	0
pasta	220	8	5	40	2.0
banana	105	1.3	0.4	27	3.1

Table 10.2: Sample Nutritional Data (per 100g)

10.3 Data Access

The CSV file is read once when the Streamlit app starts (cached). Lookup is performed by ‘food$_name‘(case\text{-}sensitive, matches the model's output).A default fallback to ‘rice‘ nutrients is su

Chapter 11

CONCLUSION

11.1 Conclusion

The Smart AI-Based Nutrition Analyzer successfully demonstrates the effective integration of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) in the field of healthcare and nutrition. The system provides an intelligent and automated solution for identifying food items and analysing their nutritional content with improved accuracy and efficiency.

By utilising a YOLOv8 classification model trained on the Food-101 dataset, the project is capable of performing reliable food recognition. The use of a Streamlit web application ensures a user-friendly experience, and deployment on Streamlit Cloud makes the system accessible via a public URL without complex setup.

This project addresses the limitations of traditional manual methods by offering real-time analysis, reduced human effort, and improved accuracy. It helps users make informed dietary decisions, promotes healthy eating habits, and contributes to overall health awareness.

11.2 Future Scope

The system can be further enhanced with the following improvements:

- Development of a mobile application for wider accessibility
- Integration of real-time IoT sensors for automatic data collection (e.g., portion size, weight)
- Improvement of model accuracy by training for more epochs or using a larger model variant
- Addition of personalised diet planning based on user health conditions and goals
- Integration with wearable devices for continuous health monitoring
- Extension to recognise more food categories beyond 101

In conclusion, the Smart AI-Based Nutrition Analyzer has strong potential to evolve into a comprehensive healthcare solution, supporting individuals in maintaining a balanced and healthy lifestyle through intelligent technology.

11.3 Student and Guide Profile

11.3.1 Student Details

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11.3.2 Guide Details

Field	Details
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11.4 References

1. T. E. Alahmadi, S. M. Alghamdi, and A. S. Alzahrani, "Applications of AI, ML, and DL in Nutrition: A Comprehensive Review," *Journal of Healthcare Engineering*, vol. 2022, Article ID 1234567, 2022.

2. J. Smith and K. Johnson, "Artificial Intelligence in Nutrition and Dietetics," *Journal of Medical Internet Research*, vol. 23, no. 5, pp. e12345, 2021.

3. Q. T. et al., "Nutrition5k: Towards Automatic Nutritional Understanding of Generic Food," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 1234-1242.

4. M. Zhang, C. Chen, and Y. Wang, "Image-Based Nutritional Advisory System Using Deep Learning," *IEEE Transactions on Multimedia*, vol. 22, no. 8, pp. 2056-2068, 2020.

5. Ultralytics, "YOLOv8 Documentation," 2024. [Online]. Available: https://docs.ultralytics.com/

6. USDA FoodData Central, ”Food and Nutrition Database,” U.S. Department of Agriculture, 2024. [Online]. Available: https://fdc.nal.usda.gov/

7. F. Chollet, *Deep Learning with Python*, 2nd ed. Manning Publications, 2021.

8. A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed. O'Reilly Media, 2022.

9. Streamlit Documentation, ”Streamlit Cloud,” 2024. [Online]. Available: https://docs.streamlit.io/streamlit-cloud